



BITS Pilani
Pilani Campus

Machine Learning

AIML CZG565

Ensemble Learning

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Disclaimer and Acknowledgement



- These content of modules & context under topics are planned by the course owner Dr. Sugata, with grateful acknowledgement to many others who made their course materials freely available online
- We here by acknowledge all the contributors for their material and inputs.
- We have provided source information wherever necessary
- Students are requested to refer to the textbook w.r.t detailed content of the presentation deck shared over canvas
- We have reduced the slides from canvas and modified the content flow to suit the requirements of the course and for ease of class presentation

Slide Source / Preparation / Review:

From BITS Pilani WILP: Profs.Sugata, Chetana, Monali, Rajavadhana, Seetha, Anita

External: CS109 and CS229 Stanford lecture notes, Dr.Andrew NG and many others who made their course materials freely available online

Course Plan

M1	Introduction & Mathematical Preliminaries
M2	Machine Learning Workflow
M3	Linear Models for Regression
M4	Linear Models for Classification ✓
M5	Decision Tree ✓
M6	Instance Based Learning ✓
M7	Support Vector Machine
M8	Bayesian Learning ✓
M9	Ensemble Learning
M10	Unsupervised Learning
M11	Machine Learning Model Evaluation/Comparison

Predictors
Estimators

Ensemble Learning

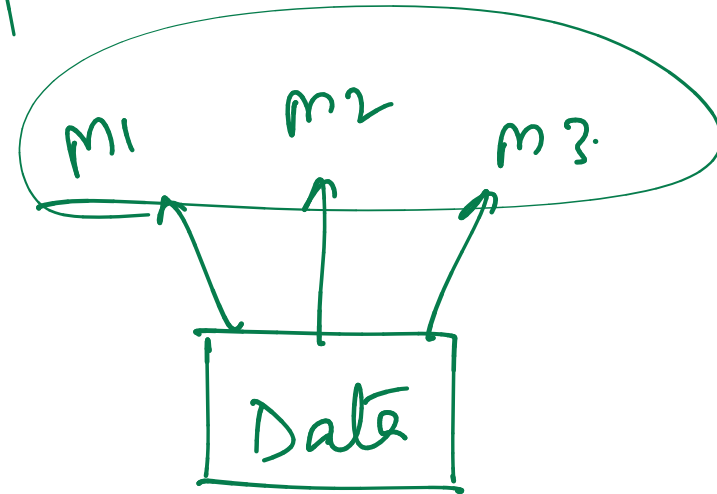
The Single Model Philosophy

- Motivation: Occam's Razor
 - “one should not increase, beyond what is necessary, the number of entities required to explain anything”
- Infinitely many models can explain any given dataset
 - Might as well pick the smallest one...

Ensemble Philosophy

- No Free Lunch Theorem: There is no algorithm that is always the most accurate
- Each learning algorithm dictates a certain model that comes with a set of assumptions
 - Each algorithm converges to a different solution and fails under different circumstances
 - The best tuned learners could miss some examples and there could be other learners which works better on (may be only) those !
 - In the absence of a single expert (a superior model) , a committee (combinations of models) can do better !
 - A committee can work in many ways ...

$E_r / Acc = 50\%$

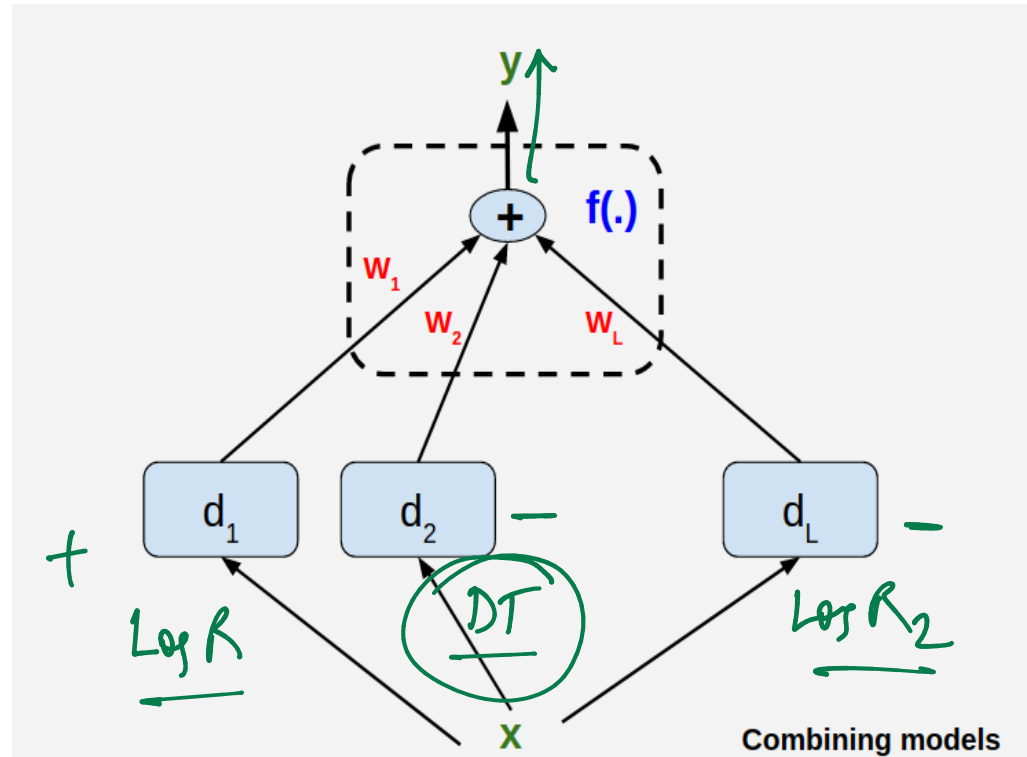


Weak learner does only slightly better than random guessing

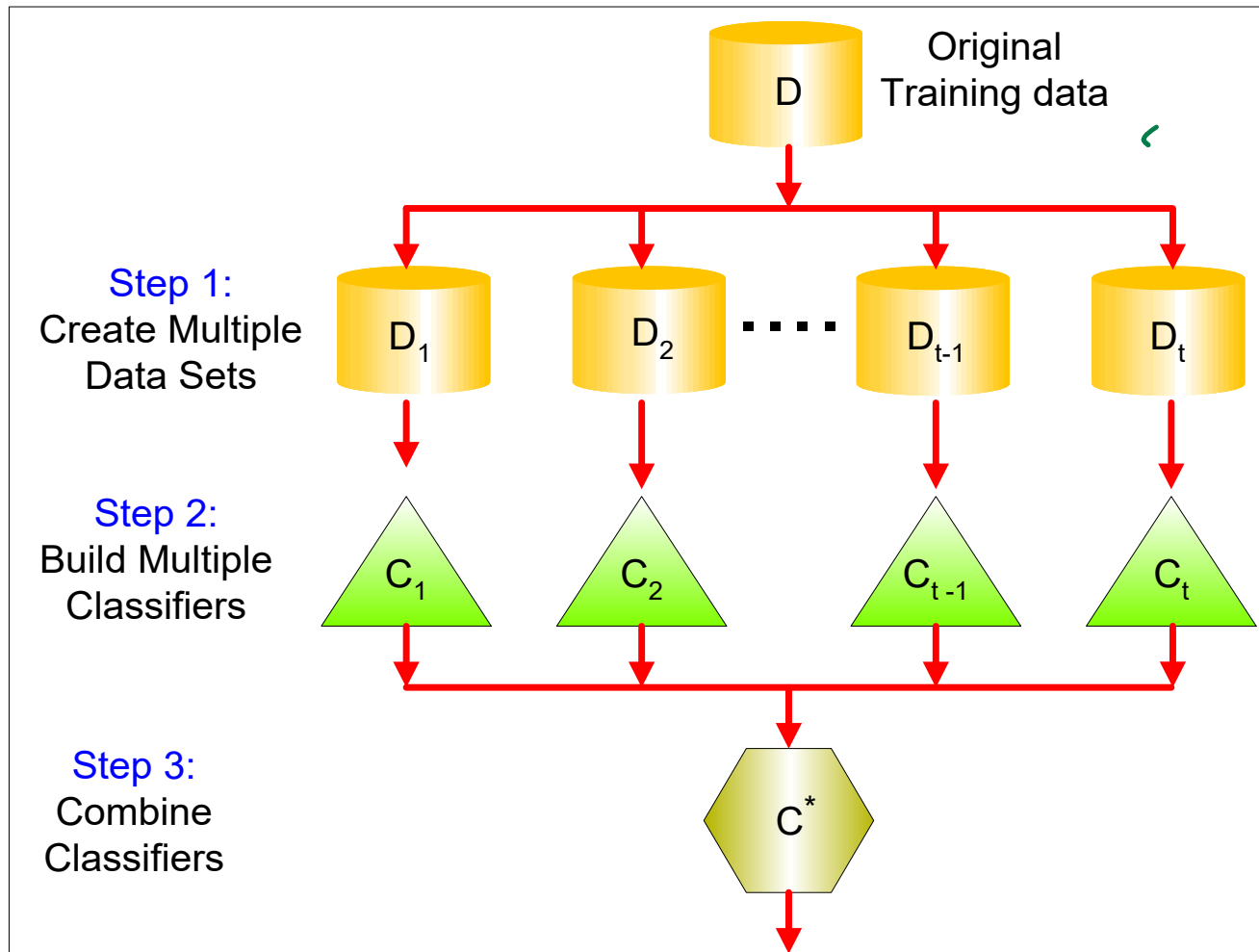
Strong diversity
Learn

Committee of Models

- Committee Members are base learners !
- Major challenges dealing with this committee
 - Expertise of each of the members (Does it help / not?)
 - Combining the results from the members for better performance

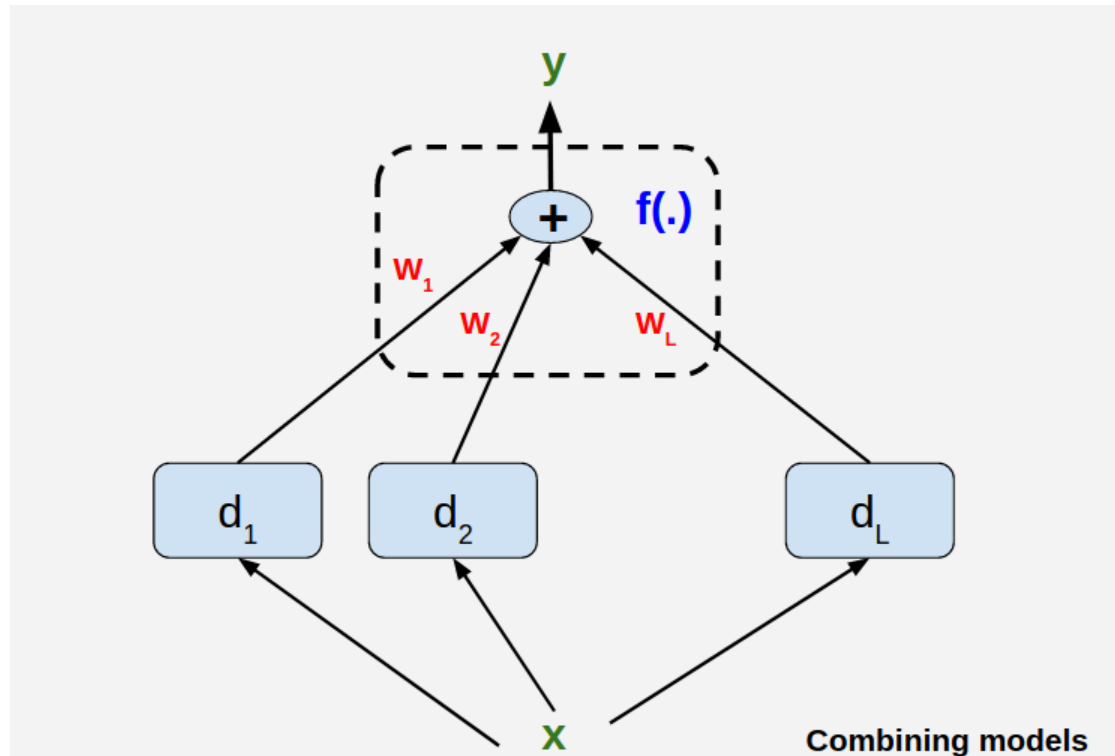


General Approach



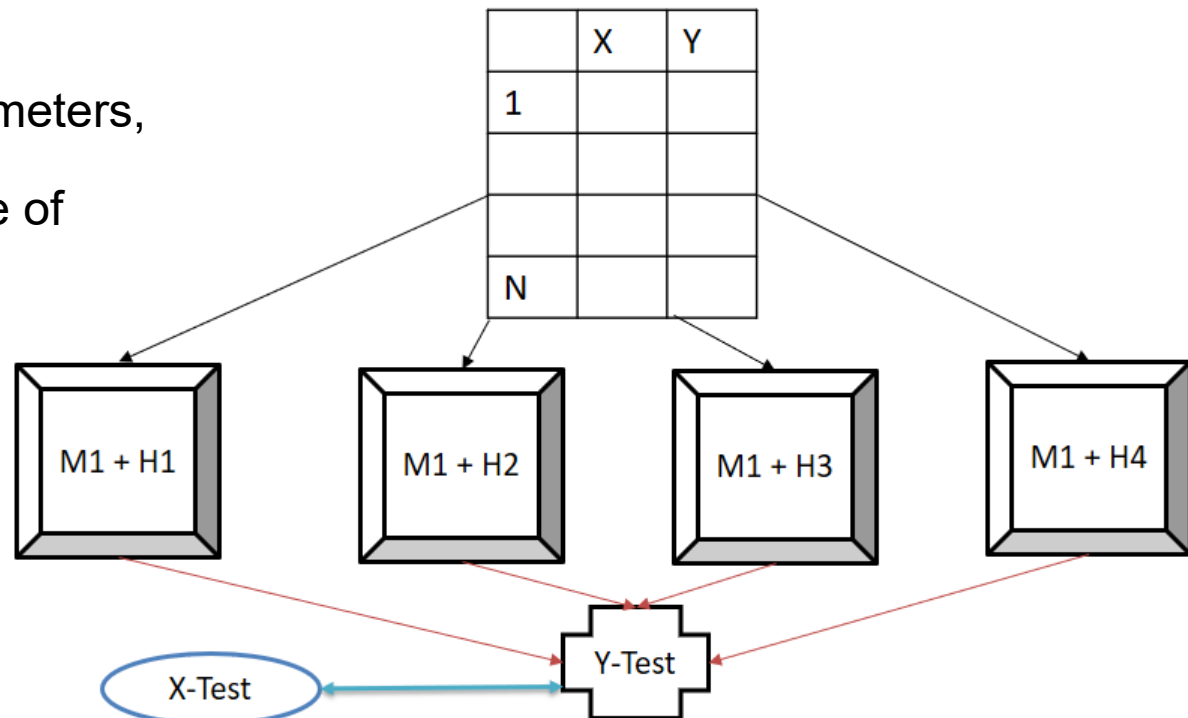
Issue 1 : On the members (Base Learners)

- It does not help if all learners are good/bad at roughly same thing
 - Need Diverse Learners



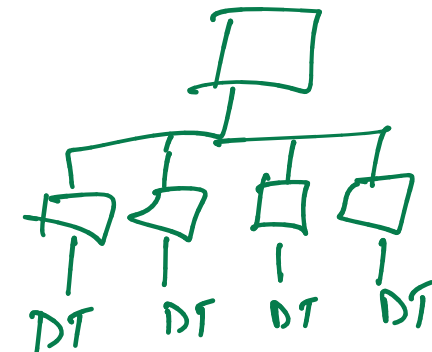
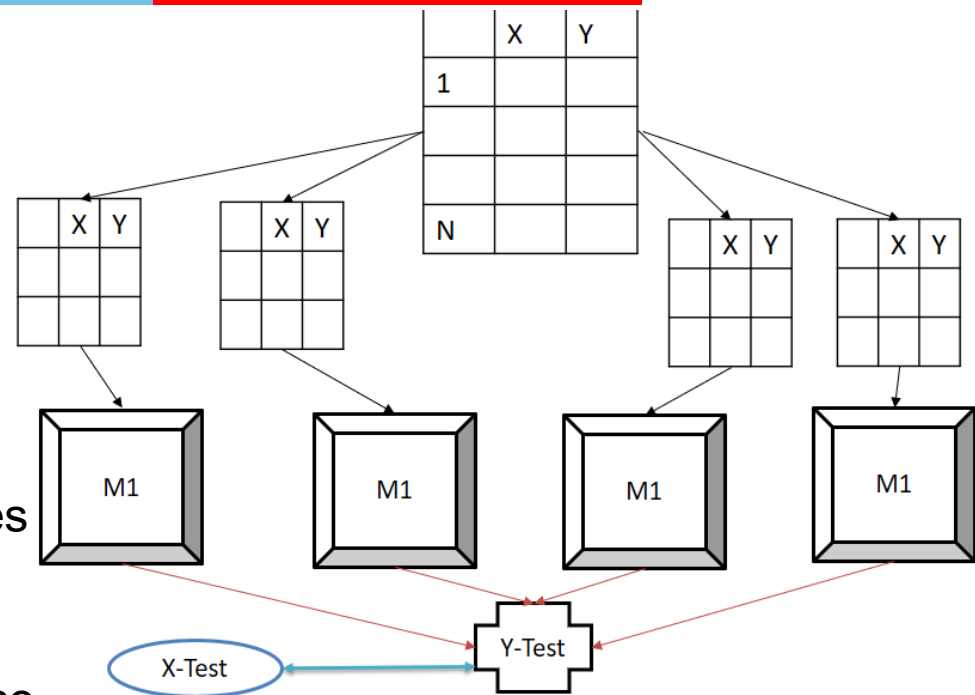
Issue 1 : On the members (Base Learners)

- Use Different Algorithms
 - ✓ – Different algorithms make different assumptions
- Use Different Hyper parameters,
 - ✓ – E.g. vary the structure of neural nets



Issue 1 : On the members (Base Learners)

- Different input representations
 - Uttered words + video information of speakers clips
 - image + text annotations
- Different training sets
 - Draw different random samples of data
 - Partition data in the input space and have learners specialized in those spaces (mixture of experts)



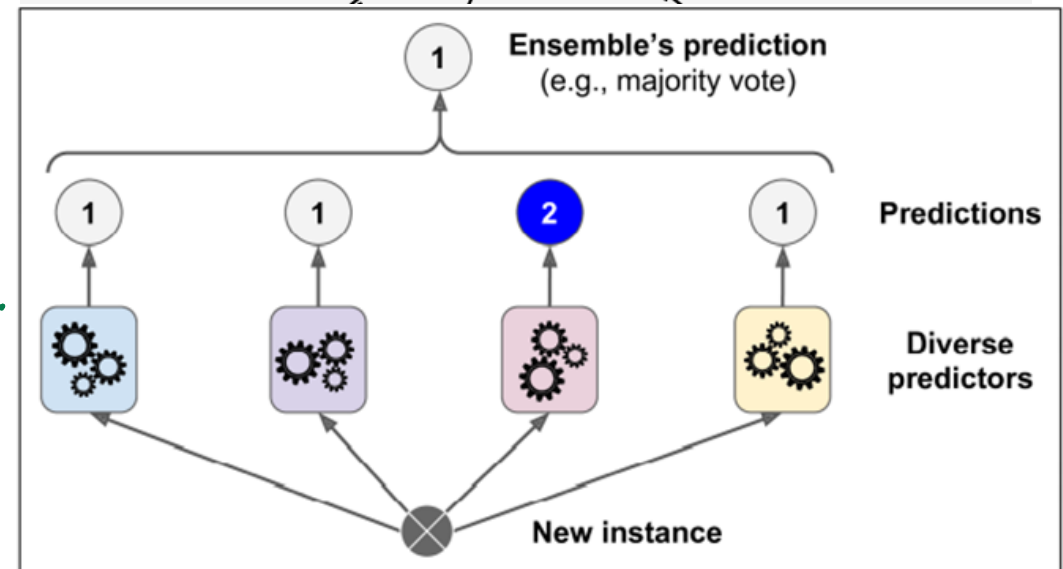
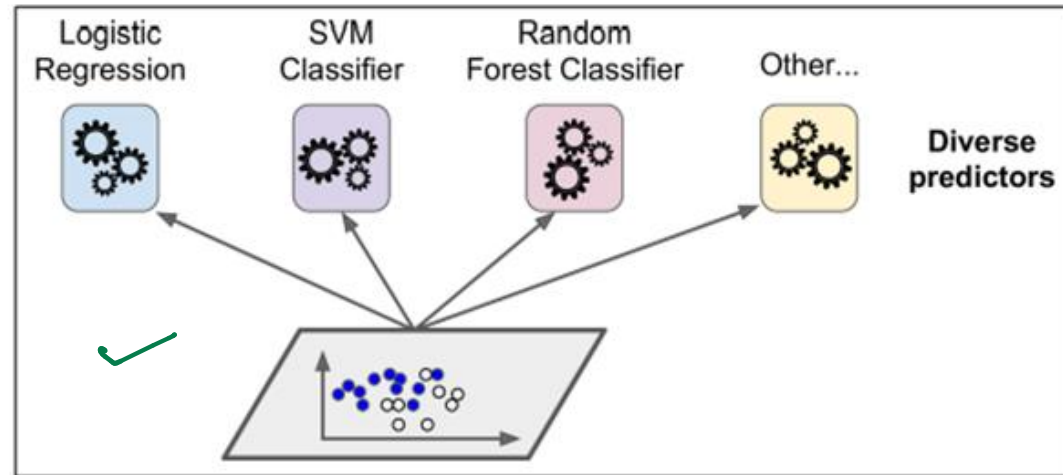
Issue -2 : Combining Results Base Learners

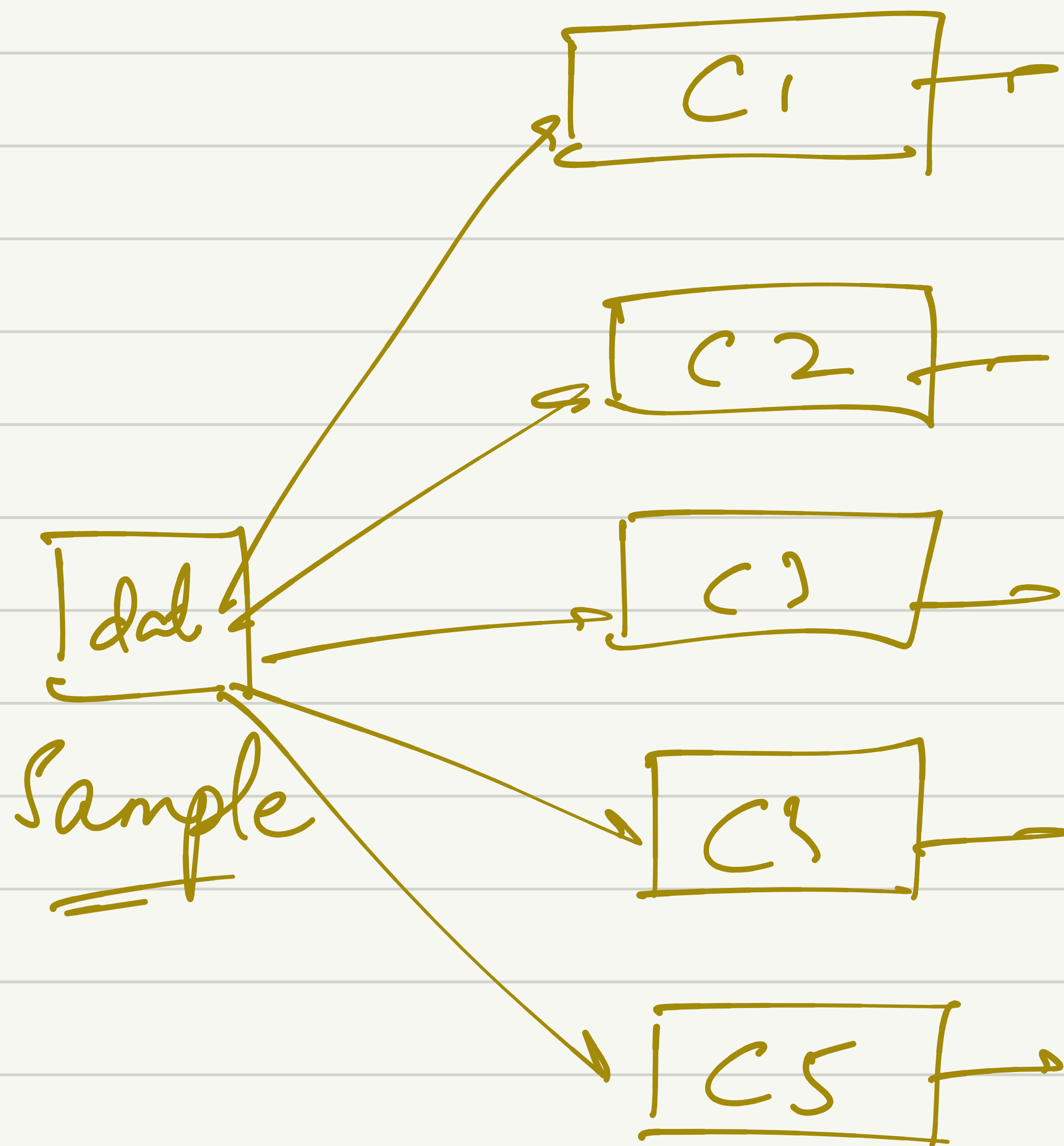
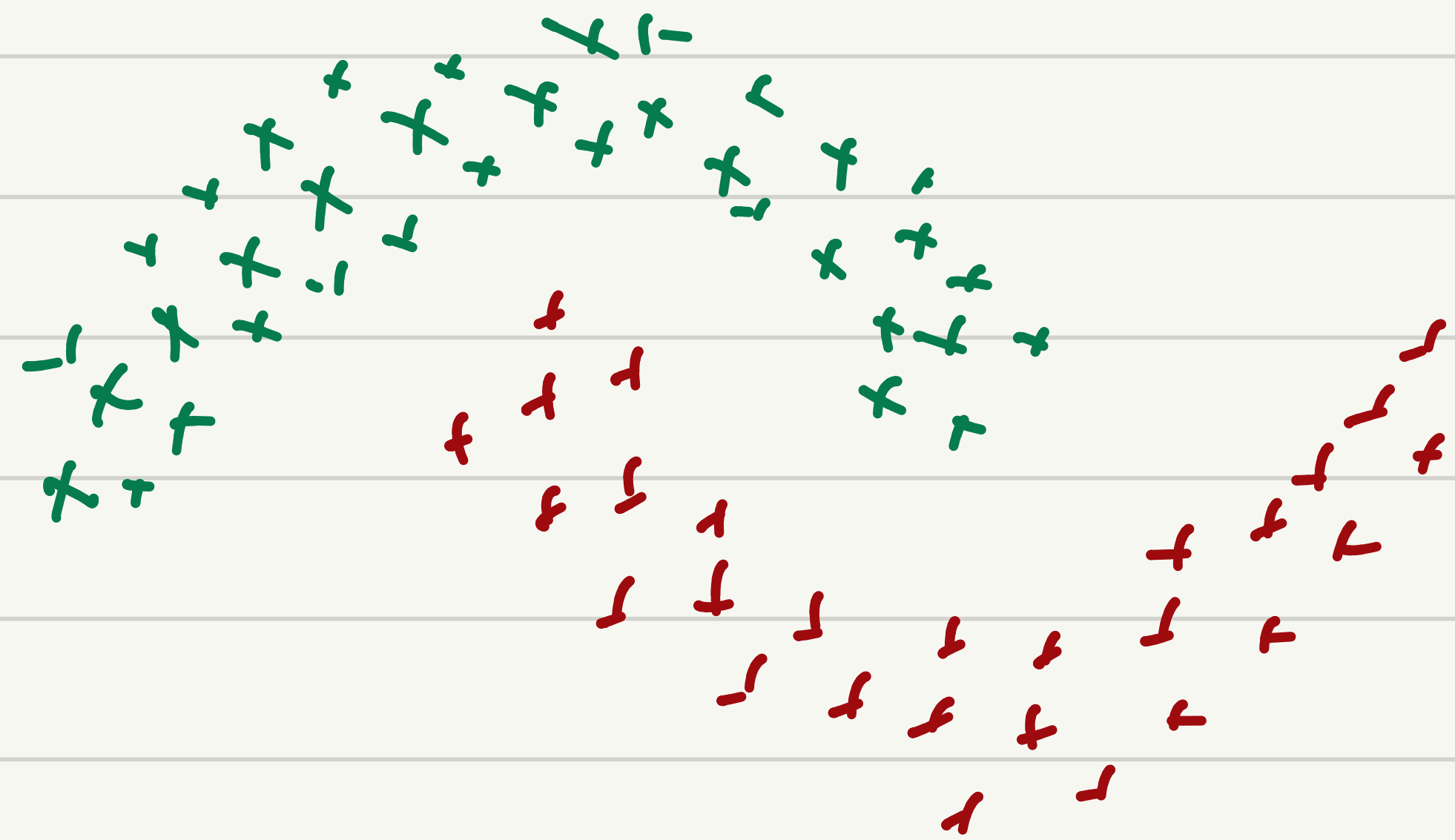
$$y = f(d_1, d_2, \dots, d_L | \Phi)$$

A Simple Combination Scheme:

$$y = \sum_{j=1}^L w_j d_j$$

$$w_j \geq 0 \text{ and } \sum_{j=1}^L w_j = 1$$

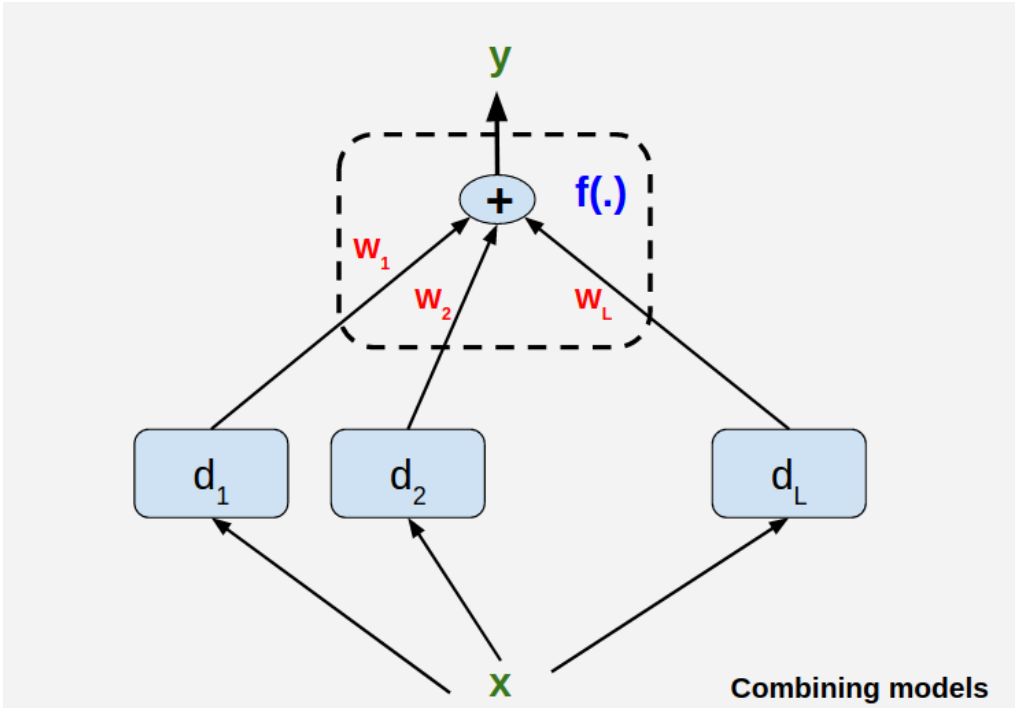




Issue -2 : Combining Results Base Learners

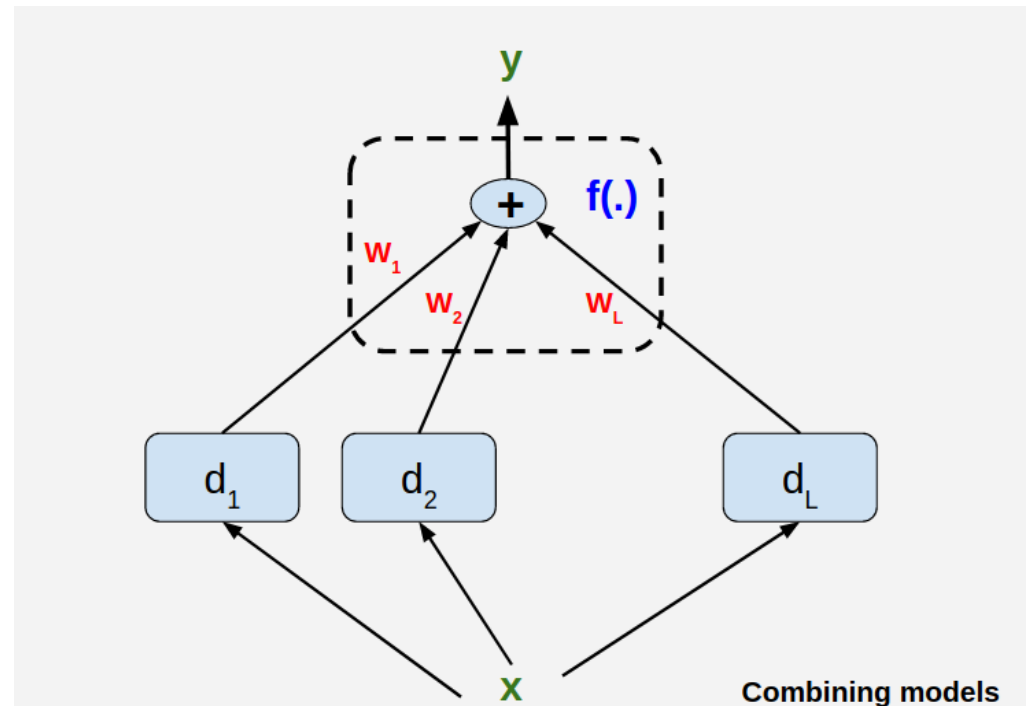
$$y = f(d_1, d_2, \dots, d_L | \Phi)$$

Rule	Fusion function $f(\cdot)$
Sum ✓	$y_i = \frac{1}{L} \sum_{j=1}^L d_{ji}$
Weighted sum ✓	$y_i = \sum_j w_j d_{ji}, w_j \geq 0, \sum_j w_j = 1$
Median ✓	$y_i = \text{median}_j d_{ji}$
Minimum ✓	$y_i = \min_j d_{ji}$
Maximum ✓	$y_i = \max_j d_{ji}$
Product ✓	$y_i = \prod_j d_{ji}$



Issue -2 : Combining Results Base Learners

	C_1	C_2	C_3
d_1	0.2	0.5	0.3
d_2	0.0	0.6	0.4
d_3	0.4	0.4	0.2
Sum Avg	0.2	0.5	0.3
Median	0.2	0.5	0.4
Minimum	0.0	0.4	0.2
Maximum	0.4	0.6	0.4
Product	0.0	0.12	0.032



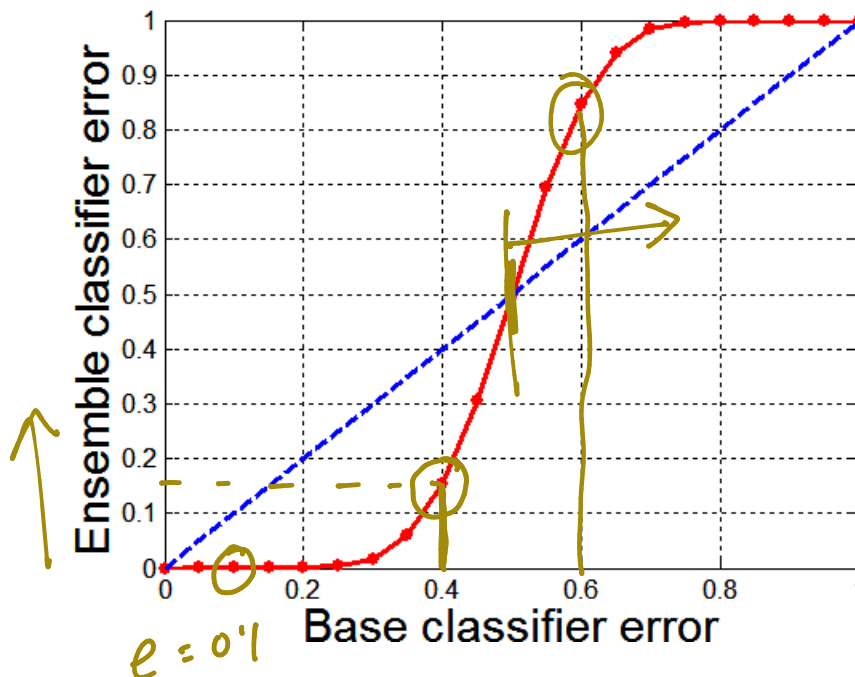
When Ensemble Methods Work?

- Ensemble classifier performs better than the base classifiers when e is smaller than 0.5
- Necessary conditions for an ensemble classifier to perform better than a single classifier:
 - Base classifiers should be independent of each other
 - ✓ – Base classifiers should do better than a classifier that performs random guessing

Necessary Conditions for Ensemble Methods

- Ensemble Methods work better than a single base classifier if:
 - All base classifiers are independent of each other
 - All base classifiers perform better than random guessing (error rate < 0.5 for binary classification)

base	err
0.4	0.15
0.1	0
0.6	0.85



$e = 0.5$ $\rightarrow$ random guess

Classification error for an ensemble of 25 base classifiers, assuming their errors are uncorrelated.

Weak
 $e = 0.4$ $e = 0.6$

→ $e = 0.4$ → weak learner.

→ $e = 0.1$ → Strong. learner

$$\text{error rate} = \frac{\#IC}{T+1C} = 50\%$$

$$\boxed{e < 50\%} = \frac{\#C(\text{better})}{T\#C}$$

learn ← $\#C > \#IC$ →

Types of Ensemble Methods

- By manipulating training set *→ instance.*
 - Example: bagging, boosting, random forests
- By manipulating input features *→ column*
 - Example: random forests

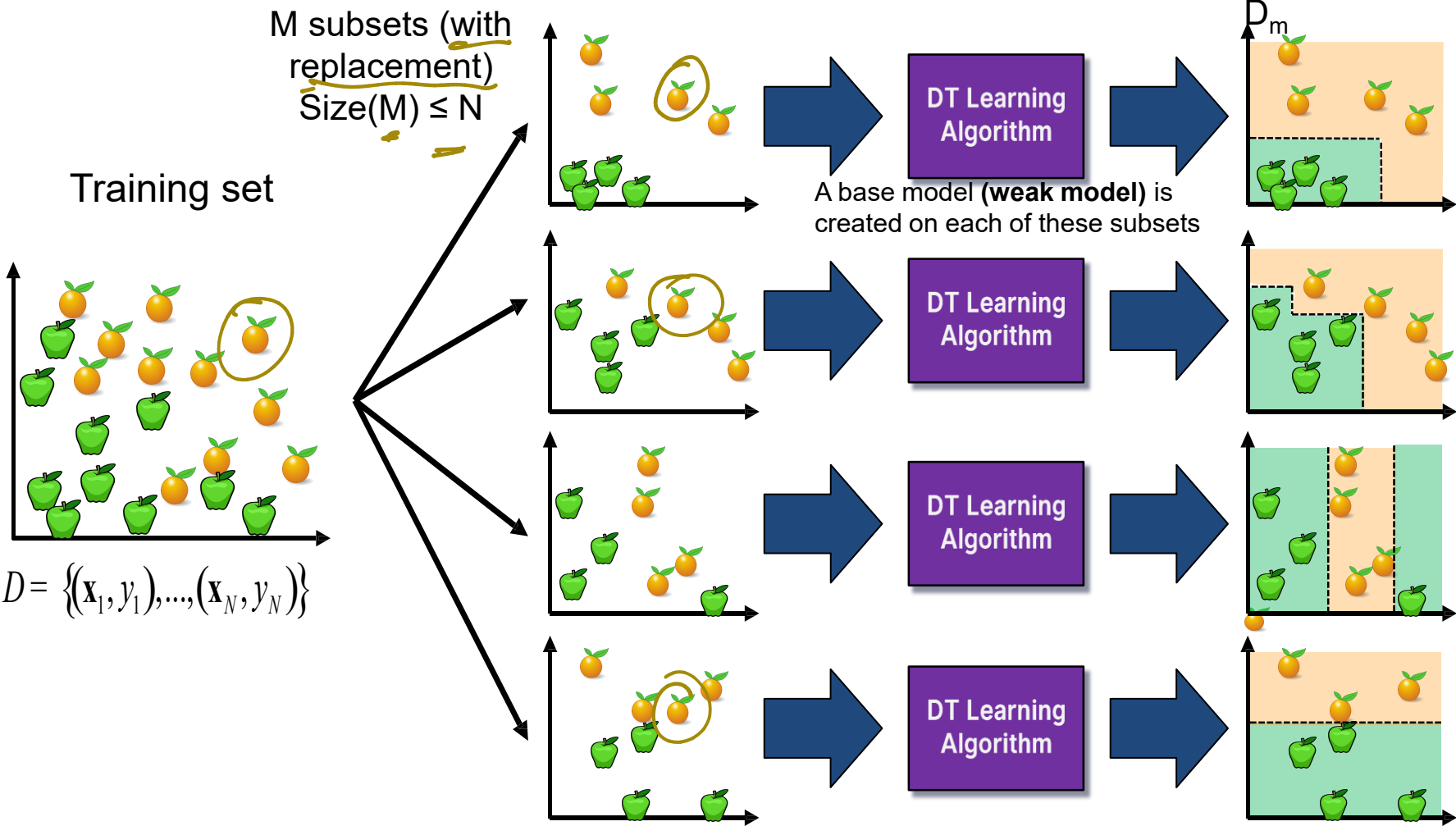
① Bagging

Bootstrap Aggregating

Bagging at training time

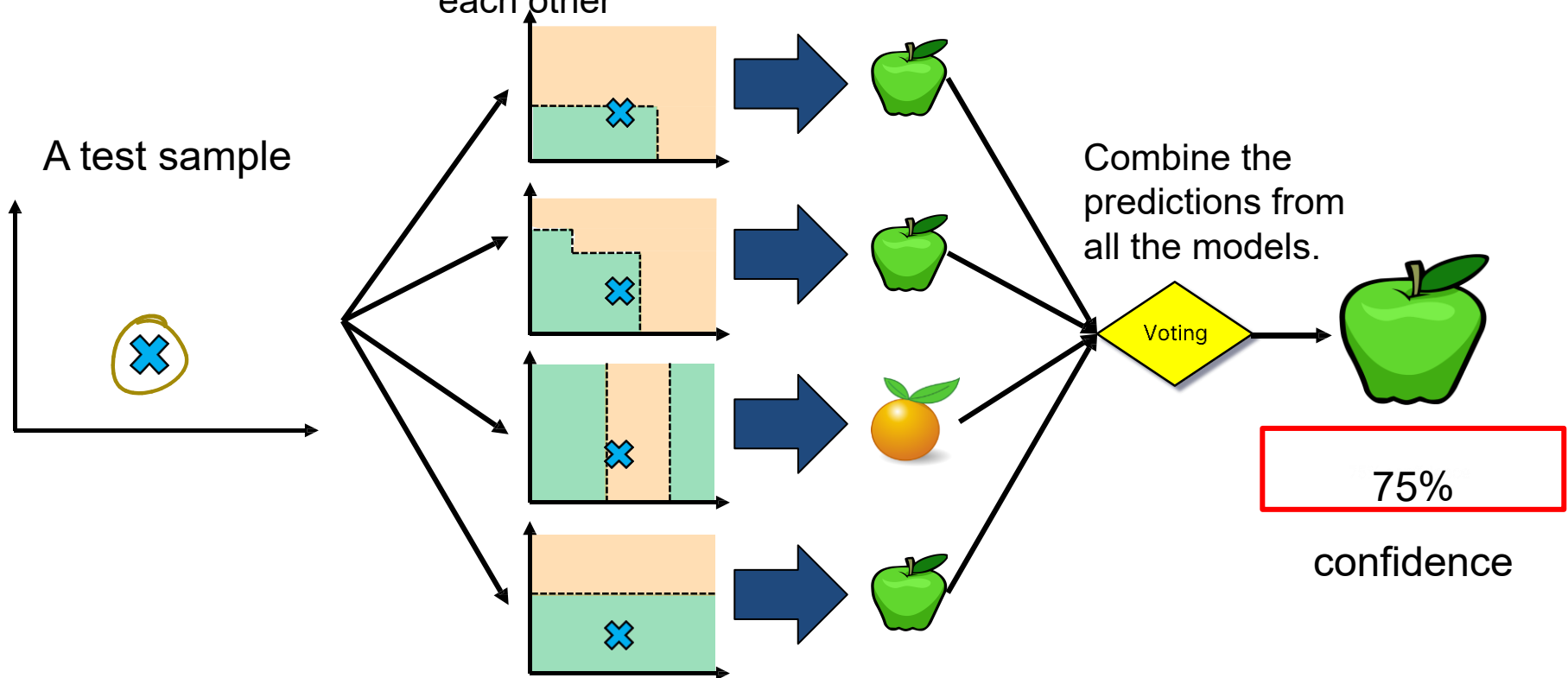
For $m=1:M$, Obtain bootstrap sample D_m from the training data D

Build a model $G_m(x)$ from bootstrap data



Bagging at inference time

The models run in **parallel** and are independent of each other



The Bagging Model

- Regression

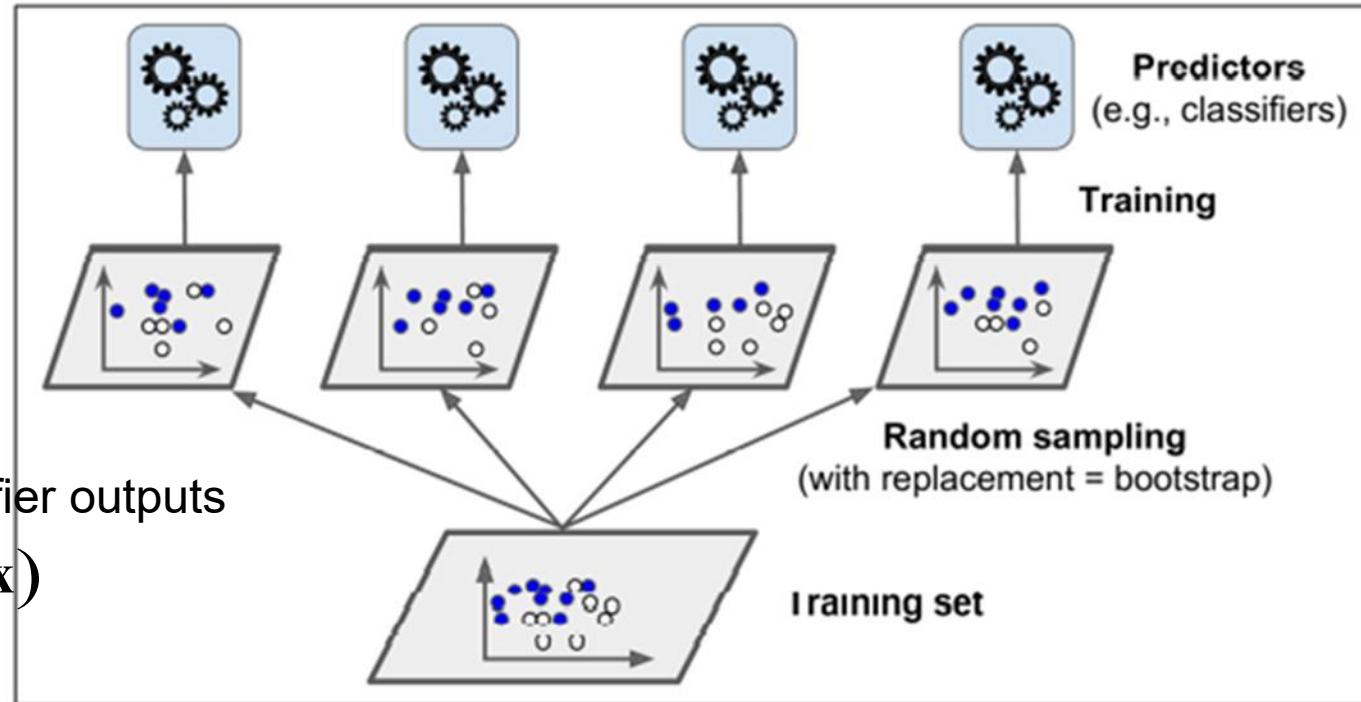
$$\hat{y} = \frac{1}{M} \sum_{m=1}^M G_m(\mathbf{x})$$



- Classification:

- Vote over classifier outputs

$$G_1(\mathbf{x}), \dots, G_M(\mathbf{x})$$



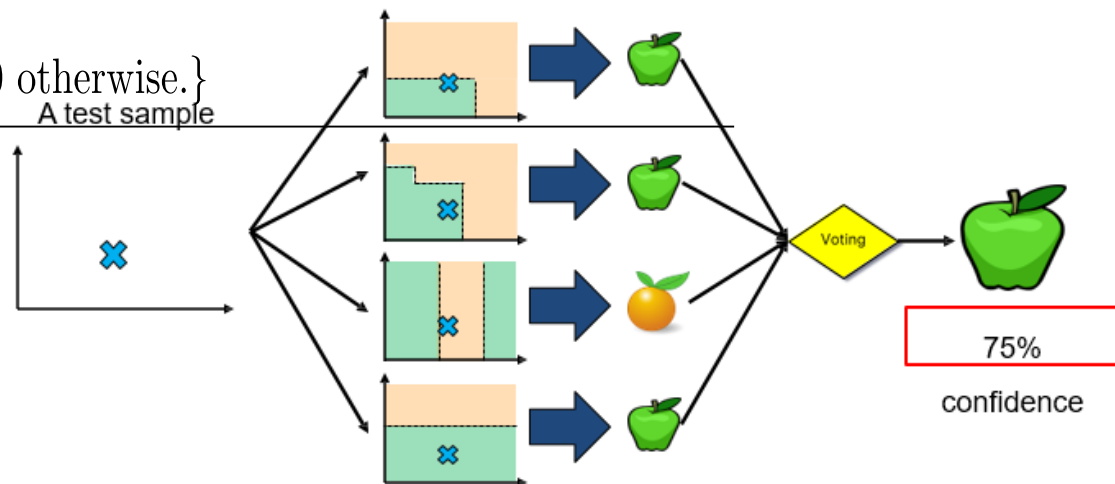
Bagging Algorithm

Algorithm 4.5 Bagging algorithm.

- 1: Let k be the number of bootstrap samples.
- 2: **for** $i = 1$ to k **do**
- 3: Create a bootstrap sample of size N , D_i .
- 4: Train a base classifier C_i on the bootstrap sample D_i .
- 5: **end for**
- 6: $C^*(x) = \operatorname{argmax}_y \sum_i \delta(C_i(x) = y).$

$\{\delta(\cdot) = 1 \text{ if its argument is true and } 0 \text{ otherwise.}\}$

A test sample



Bagging Example

Consider 1-dimensional data set:

Original Data:

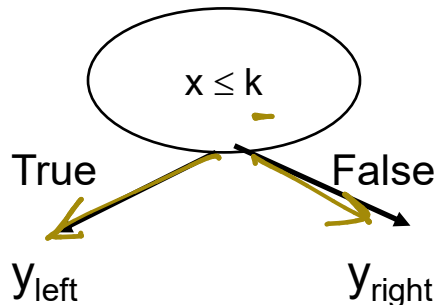
x	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1
y	1	1	1	-1	-1	-1	-1	1	1	1

Classifier is a decision stump ✕

Decision rule: $x \leq k$ versus $x > k$

$x < 0.35$ or $x \geq 0.75$.

Split point k is chosen based on entropy



Decision tree with one internal node (the root) which is immediately connected to the terminal nodes (its leaves). Decision stump makes a prediction based on the value of just a single input feature. Sometimes they are also called 1-rules

Handwritten notes: $x \leq 0.5$, $+1$, -1 , and a dollar sign with arrows pointing to the data table.

Bagging Example

innovate

achieve

lead

Bagging Round 1:

x	0.1	0.2	0.2	0.3	0.4	0.4	0.5	0.6	0.9	0.9
y	1	1	1	1	-1	-1	-1	-1	1	1

$x \leq 0.35 \rightarrow y = 1$
 $x > 0.35 \rightarrow y = -1$

Bagging Round 2:

x	0.1	0.2	0.3	0.4	0.5	0.5	0.9	1	1	1
y	1	1	1	-1	-1	-1	1	1	1	1

$x \leq 0.7 \rightarrow y = 1$ ✓
 $x > 0.7 \rightarrow y = -1$

Bagging Round 3:

x	0.1	0.2	0.3	0.4	0.4	0.5	0.7	0.7	0.8	0.9
y	1	1	1	-1	-1	-1	-1	-1	1	1

$x \leq 0.35 \rightarrow y = 1$
 $x > 0.35 \rightarrow y = -1$

Bagging Round 4:

x	0.1	0.1	0.2	0.4	0.4	0.5	0.5	0.7	0.8	0.9
y	1	1	1	-1	-1	-1	-1	-1	1	1

$x \leq 0.3 \rightarrow y = 1$
 $x > 0.3 \rightarrow y = -1$

Bagging Round 5:

x	0.1	0.1	0.2	0.5	0.6	0.6	0.6	1	1	1
y	1	1	1	-1	-1	-1	-1	1	1	1

$x \leq 0.35 \rightarrow y = 1$
 $x > 0.35 \rightarrow y = -1$

Bagging Example

Bagging Round 6:

x	0.2	0.4	0.5	0.6	0.7	0.7	0.7	0.8	0.9	1
y	1	-1	-1	-1	-1	-1	-1	1	1	1

$x \leq 0.75 \rightarrow y = -1$
 $x > 0.75 \rightarrow y = 1$

Bagging Round 7:

x	0.1	0.4	0.4	0.6	0.7	0.8	0.9	0.9	0.9	1
y	1	-1	-1	-1	-1	1	1	1	1	1

$x \leq 0.75 \rightarrow y = -1$
 $x > 0.75 \rightarrow y = 1$

Bagging Round 8:

x	0.1	0.2	0.5	0.5	0.5	0.7	0.7	0.8	0.9	1
y	1	1	-1	-1	-1	-1	-1	1	1	1

$x \leq 0.75 \rightarrow y = -1$
 $x > 0.75 \rightarrow y = 1$

Bagging Round 9:

x	0.1	0.3	0.4	0.4	0.6	0.7	0.7	0.8	1	1
y	1	1	-1	-1	-1	-1	-1	1	1	1

$x \leq 0.75 \rightarrow y = -1$
 $x > 0.75 \rightarrow y = 1$

Bagging Round 10:

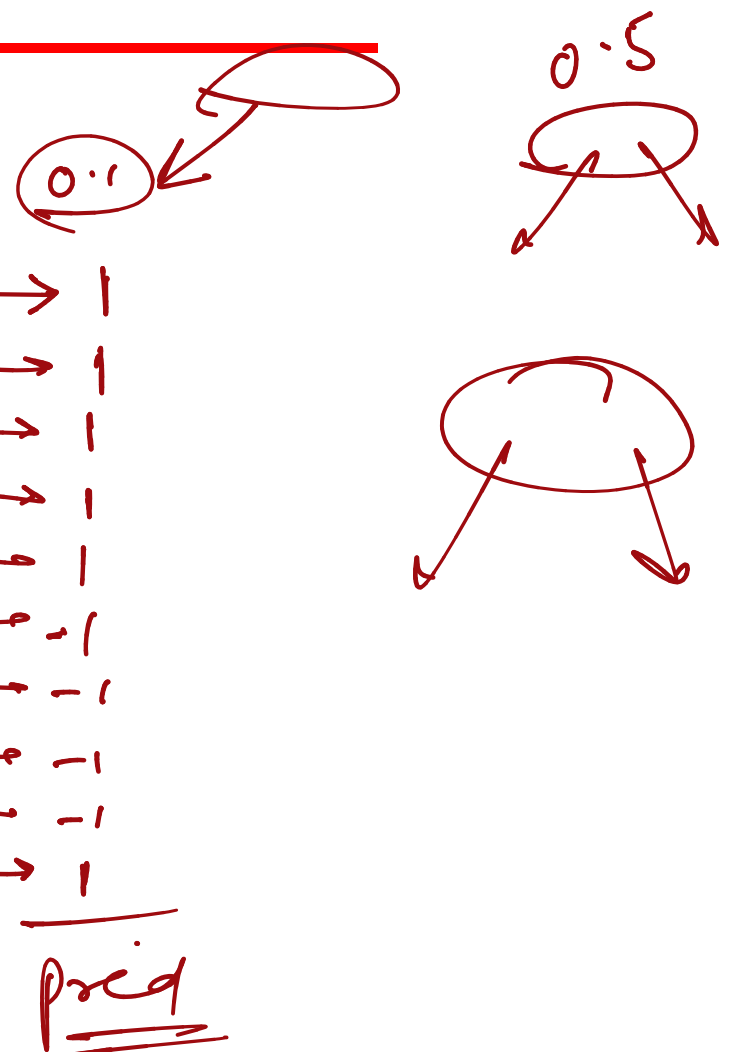
x	0.1	0.1	0.1	0.1	0.3	0.3	0.8	0.8	0.9	0.9
y	1	1	1	1	1	1	1	1	1	1

$x \leq 0.05 \rightarrow y = 1$
 $x > 0.05 \rightarrow y = 1$

Bagging Example

Summary of Training sets:

Round	Split Point	Left Class	Right Class
1	0.35 ✓	1	-1
2	0.7 ✓	1	1
3	0.35 ✓	1	-1
4	0.3 ✓	1	-1
5	0.35 ✓	1	-1
6	0.75 ✓	-1	1
7	0.75	-1	1
8	0.75	-1	1
9	0.75	-1	1
10	0.05	1	1



Bagging Example

- Assume test set is the same as the original data
- Use majority vote to determine class of ensemble classifier

Original Data:

Test

x	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1
y	1	1	1	-1	-1	-1	-1	1	1	1

Sum = Ave
Vote

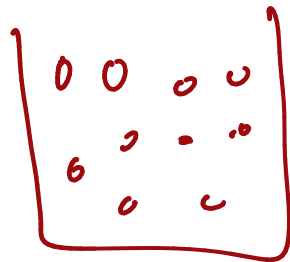
Round	x=0.1	x=0.2	x=0.3	x=0.4	x=0.5	x=0.6	x=0.7	x=0.8	x=0.9	x=1.0
1	1	1	1	-1	-1	-1	-1	-1	-1	-1
2	1	1	1	1	1	1	1	1	1	1
3	1	1	1	-1	-1	-1	-1	-1	-1	-1
4	1	1	1	-1	-1	-1	-1	-1	-1	-1
5	1	1	1	-1	-1	-1	-1	-1	-1	-1
6	-1	-1	-1	-1	-1	-1	-1	1	1	1
7	-1	-1	-1	-1	-1	-1	-1	1	1	1
8	-1	-1	-1	-1	-1	-1	-1	1	1	1
9	-1	-1	-1	-1	-1	-1	-1	1	1	1
10	1	1	1	1	1	1	1	1	1	1
Sum	2	2	2	-6	-6	-6	-6	2	2	2
Sign	1	1	1	-1	-1	-1	-1	1	1	1

Predicted Class

↑ ↑ ↑ ↑

Bagging – Effect on Variance

- Bagging helps make models like decision trees less jumpy or unstable.
- It builds many models using different random samples of the data.
- By combining their predictions, the final result becomes more consistent.
- This makes the model less likely to overfit and work better on new data.



100 people

Bagging – Effect on Bias

Left DT → high bias.

→ Bagging
X

- If a base classifier is stable, i.e., robust to minor perturbations in the training set, then the error of the ensemble is primarily caused by bias in the base classifier.
- In this situation, bagging may not be able to improve the performance of the base classifiers significantly.
- It may even degrade the classifier's performance because the effective size of each training set is about 37% smaller than the original data.

N

63%

References

- Introduction to Data Mining, by Pang-Ning Tan, Michael Steinbach, Vipin Kumar
- Bishop - Pattern Recognition And Machine Learning - Springer 2006
- https://en.wikipedia.org/wiki/Gradient_boosting#:~:text=Gradient%20boosting%20is%20a%20machine,which%20are%20typically%20decision%20trees
- <https://medium.com/nerd-for-tech/gradient-boost-for-regression-explained-6561eec192cb>